AR-10 / SR-25 / LR-308 Big-Bore Platform

ArmaLite / Knight's Armament / DPMS / Aero Precision • Large-Frame Modern Sporting Rifles (7.62x51mm NATO & .308 Winchester / 6.5 CM)

Manufacturer:	ArmaLite / Knight's Armament / DPMS / Aero Precision	**Category:**	Rifles & Carbines
Action Type:	Direct Gas Impingement Large-Frame Semi-Auto	**Era / Service:**	Military DMR & Hunting (1956–Present)
Calibers:	7.62x51mm NATO, .308 Winchester, 6.5 Creedmoor	**Covered Models:**	ArmaLite AR-10A / AR-10B, KAC SR-25 / M110 SASS, DPMS LR-308 (Gen 1 / Gen 2), Aero Precision M5

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Pattern Compatibility Check (ArmaLite vs DPMS) • Gas Port Sizing & Buffer Weight Tuning • Dual Extractor Pin Servicing • Barrel Nut Handguard Alignment • Headspace Gauging

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
PATTERN DIFFERENCES	CRITICAL ARMORER RULE: ArmaLite pattern features angular receiver cut; DPMS pattern features radius curved cut (uppers and lowers do NOT interchange)
HEADSPACE 308 WIN	GO = 1.6300", NO-GO = 1.6340", FIELD = 1.6380" (NATO chambers run longer: GO = 1.6355", FIELD = 1.6455")
DUAL EXTRACTORS	Many modern .308 bolts feature dual spring-loaded ejectors to throw heavy brass clear of oversized ejection port
GAS SYSTEM LENGTH	Rifle-length or +2" extended gas system recommended for 6.5 Creedmoor to prevent high-pressure early unlocking

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Pierced Primers / Blown Primers in Bolt Carrier High pressure round craters or punctures primer cup, blowing metal shaving into firing pin hole	• Standard large firing pin aperture (0.080") used with high pressure 6.5 Creedmoor ammunition. • Over-gassed rifle causing bolt to unlock before chamber pressure drops.	**Gunsmith Action: Install high-pressure bolt with reduced diameter firing pin tip (0.060" pin) and tight aperture to prevent primer flow. Tune gas with adjustable gas block.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
AR-10 / .308 AR Build, Tuning & Gas Port Optimization	Brownells Gunsmithing Bench	Comparing ArmaLite vs DPMS patterns, sizing gas ports for .308 Win vs 6.5 Creedmoor, and setting headspace on big-bore AR rifles.